Project 2: Soil Salinity Mapping & Crop Zoning

Satellite-Driven Salinity Prediction and Halophyte Crop Recommendation System

1. Problem Statement

Farmers and agricultural planners in coastal, arid, and irrigation-heavy regions need to know *where* soil salinity is a problem and *which crops* are viable there — but ground-truth soil testing (electrical conductivity, EC) is expensive, sparse, and slow to scale across large regions. Meanwhile, satellite imagery covering these regions is free and updated every few days.

There is no widely accessible, easy-to-use tool that combines (a) satellite-derived salinity prediction with (b) an actionable crop recommendation layer specifically oriented around salt-tolerant/halophyte crops. Existing soil salinity research largely stops at "here is a map of predicted EC" without connecting it to a decision layer a farmer or planner could act on.

Core question: For a given region, can we predict soil salinity from freely available satellite imagery, and automatically recommend which salt-tolerant crop (or standard crop) is agronomically appropriate for each zone?

2. Objectives

- Build an ML model that predicts soil salinity (EC or a salinity index class) from multispectral satellite data.
- Validate against published ground-truth EC datasets.
- Layer a rule-based + ML crop recommendation system on top of the salinity prediction.
- Deliver an interactive map output for a chosen pilot region.

3. Solution Methodology

Phase 1 — Define Pilot Region & Gather Ground Truth

1. Choose a pilot region with known salinity issues and published ground-truth EC data available in literature/open datasets — good candidates: Nile Delta (Egypt), Indus Basin (Pakistan), San Joaquin Valley (California), or coastal Bangladesh.
2. Collect any available point-based EC measurements from these studies (soil survey reports, supplementary data tables).
3. Supplement with FAO/ISRIC global salinity layers as a coarse baseline.

Phase 2 — Satellite Data Acquisition

1. Use **Google Earth Engine (GEE)** to pull Sentinel-2 (10m resolution, ideal) and/or Landsat 8/9 imagery for the pilot region and time period matching your ground-truth data.
2. Compute standard salinity-relevant spectral indices directly in GEE:
   - **NDSI** (Normalized Difference Salinity Index)
   - **SI** (Salinity Index) and SI-T variants
   - **BI** (Brightness Index)
   - **NDVI** (vegetation health proxy — salinity suppresses vegetation vigor)
3. Export a stacked raster of these indices + raw bands for the region.

Phase 3 — Model Training

1. Extract pixel values at ground-truth EC point locations to build a training table: (spectral indices + bands) → EC value.
2. Train regression models:
   - *Baseline:* Random Forest Regressor / Gradient Boosting (XGBoost) — strong performance on tabular spectral data.
   - *Advanced:* a small CNN operating on image patches to capture spatial context (if 50+ ground-truth points are available).
3. Validate with k-fold cross-validation; report RMSE against held-out ground truth.
4. Apply the trained model across the full pilot region raster to generate a continuous predicted-EC map.

Phase 4 — Crop Recommendation Layer

1. Build a lookup table of crop salinity tolerance thresholds (EC in dS/m) from established agronomic literature (e.g., FAO tables):
   - **Sensitive:** rice, beans (~1-3 dS/m)
   - **Moderately tolerant:** wheat, sorghum (~4-6 dS/m)
   - **Tolerant:** barley, cotton (~6-8 dS/m)
   - **Halophyte/highly tolerant:** quinoa, *Salicornia*, *Atriplex*, date palm, Salvadora (8+ dS/m)
2. For each pixel/zone in the predicted-EC map, assign the highest-value crop whose tolerance threshold is not exceeded.
3. Optionally add a simple economic weighting (rough market price per crop) to recommend the most profitable feasible crop.

Phase 5 — Output & Visualization

- Interactive map (using Folium, Leaflet, or GEE's apps) showing:
  - Predicted salinity layer
  - Recommended crop overlay

    - Confidence/uncertainty layer
- Optional: simple web app (Streamlit) where a user clicks a location and gets "predicted EC: X dS/m → recommended crop: Y."

4. Datasets & Resources

Resource	Purpose	Access
Google Earth Engine	Sentinel-2/Landsat imagery + cloud compute	earthengine.google.com
Copernicus Open Access Hub	Direct Sentinel-2 imagery download	dataspace.copernicus.eu
USGS EarthExplorer	Landsat imagery	earthexplorer.usgs.gov
ISRIC SoilGrids	Global soil property baseline layers (texture, carbon)	soilgrids.org
FAO Harmonized World Soil Database	Coarse global soil salinity context	fao.org/soils-portal
Published EC datasets	Model calibration (Zenodo, Dryad, Mendeley Data)	zenodo.org, datadryad.org
FAO crop salt-tolerance tables	Crop recommendation thresholds (Ayers & Westcot)	fao.org

5. Tools & Tech Stack

- **Google Earth Engine (JS/Python API):** imagery access + index computation at scale
- **Python:** geemap, rasterio, geopandas for geospatial processing
- **ML:** scikit-learn (Random Forest, Gradient Boosting), XGBoost; PyTorch/TensorFlow for CNN
- **Visualization:** Folium/Leaflet for interactive maps, Streamlit for front-end
- **Compute:** Free GEE + Google Colab (no paid infrastructure required)

6. Suggested Timeline

Weeks	Milestone
1	Pilot region selection + ground-truth data collection
2–3	Satellite data pull + spectral index computation via GEE

Weeks	Milestone
4–5	Model training + validation against ground truth
6	Full-region salinity map generation
7	Crop recommendation layer + economic weighting
8	Interactive map/app build + final writeup

7. Deliverables

1. Trained salinity prediction model + validation metrics (RMSE, R^2) against ground truth.
2. Full-region predicted salinity map (raster + visualized).
3. Crop recommendation overlay map.
4. Interactive dashboard/app.
5. Written methodology report — usable as a portfolio piece or a starting point for an actual pilot deployment.

8. Key Risks & Mitigations

- **Risk:** Sparse or hard-to-find ground-truth EC data for calibration.
  Mitigation: start with a region known to have published soil surveys (Nile Delta and Indus Basin); as a fallback, use FAO/ISRIC coarse salinity classes as weak labels.
- **Risk:** Spectral indices are confounded by other factors (moisture, crop type, bare soil vs. vegetated).
  Mitigation: include NDVI and seasonal composites (multiple dates) to disentangle salinity signal from moisture/vegetation noise.
- **Risk:** Model may not generalize outside the pilot region.
  Mitigation: clearly scope deliverable as region-specific; note in writeup that transferability requires new calibration.